Supplementary Material

The Non-Canonical Frontier in Peptide-MHC-I Binding Prediction: A Benchmark-Hygiene Case Study

Supplementary Note S — Molecular Dynamics of the K-E63 Salt Bridge

This note reports the molecular-dynamics (MD) analysis referenced in the main text (§2.4), which is deliberately excluded from the primary structural argument and presented here for completeness.

S.1 Methods

The KRAS G12V neoepitope YKLVVVGAV was modelled into the HLA-A*02:01 peptide-binding groove using the 1DUZ template (template peptide LLFGYPVYV replaced in silico). The complex was solvated and equilibrated, then simulated for 100 ns with OpenMM 8.5.2 and the amber14 force field. Three independent replicas (seeds 42 and 99) were run to assess convergence. MM-GBSA binding energies were computed over the final 50 ns (200 frames).

S.2 Results

System stability. The complex remained stable over 100 ns: the MHC α-chain, β2-microglobulin, and peptide backbone RMSDs converged to 0.15, 0.09, and 0.07 Å, respectively (Fig. S-MD2). Whole-complex backbone RMSD reached 6.1 Å, reflecting flexible regions outside the peptide-binding groove.

The K-E63 salt bridge is transient, not persistent. The P2-Lys–Glu63 minimum distance fluctuated between 3.6 and 5.9 Å across the three replicas (mean $\approx$ 4.1 Å), with transient close contacts ($<$ 3 Å) in $\sim$14% of frames in the primary trajectory (Fig. S-MD3; Fig. S-MD1). The salt bridge therefore forms intermittently but does not persist as a stable interaction over the simulated timescale.

Binding energy. MM-GBSA over the final 50 ns gave $\Delta G = -19.6 \pm 46.3$ kcal/mol (mean $\pm$ SD, n = 200), dominated by gas-phase electrostatics ($\Delta E = -19.5$ kcal/mol) with a negligible solvation term. The large variance precludes a precise binding estimate (Fig. S-MD4).

Hydrogen bonding. 3,574 unique hydrogen bonds were observed across the trajectory, with 14 persistent pairs (occupancy $>$ 30%).

S.3 Interpretation

The MD is consistent with the conservative framing in the main text (§3.5): a charged P2-Lys can transiently reach Glu63, but the K-E63 salt bridge is not a stable, persistent interaction over the

simulated timescale. These data support treating the salt-bridge hypothesis as qualitative, pending a more rigorous free-energy calculation (e.g., alchemical transformation or umbrella sampling) and experimental validation (e.g., X-ray crystallography or cryo-EM of the YKLVVVGAV–HLA-A*02:01 complex).

Supplementary Tables — Findings Summary

All numbers are validated, cross-predictor, seed-robust, temporal-holdout-checked.

Table S1. Predictor comparison — canonical vs Type B sensitivity (pooled)

Predictor class	Canonical sens	Type B (non-canonical) sens	n
Deep FFN (sequence)	0.979	0.607	14,488
MHCflurry (sequence)	0.986	0.668	14,488
MUNIS (structure)	0.950	0.647	6,224
Takeaway	∼0.98 solved	∼0.65 plateau (all classes)	

Table S2. Pan16 — 15-allele panel, per-allele non-canonical sensitivity

Allele	sens@0.95	Allele	sens@0.95
HLA-A*02:01	0.880	HLA-B*08:01	0.880
HLA-A*01:01	0.800	HLA-B*15:01	0.960
HLA-A*03:01	**1.000**	HLA-B*27:05	0.080
HLA-B*07:02	0.640	HLA-B*35:01	0.840
HLA-B*44:03	**1.000**	HLA-B*40:01	**1.000**
HLA-A*11:01	**1.000**	HLA-B*49:01	**1.000**
HLA-A*24:02	0.760	HLA-B*51:01	0.520
		HLA-B*57:01	0.920
POOLED (15 alleles)	**0.819** (307/375)		

Weak tail = thin-training alleles (B*27:05 n=150, B*51:01 n=1,665) — data scarcity, not architecture.

Table S3. Temporal holdout — generalization to post-2022 epitopes

Allele	pre-2021 sens@0.95	post-2022 sens@0.95	Delta
HLA-A*02:01	0.879 (n=2704)	0.824 (n=1151)	−0.055
HLA-A*03:01	0.905 (n=1744)	0.936 (n=803)	**+0.031**
HLA-B*07:02	0.898 (n=2315)	0.973 (n=865)	**+0.075**

Post-2022 epitopes were published after MHCflurry's training cutoff (unseen by the labeler). Preserved/improved sensitivity = NOT overfit to the IEDB era.

Table S4. P2 virtual mutagenesis — sidechain reach to Glu63

K-E63 salt bridge is one instance of a general principle: positively charged P2 residues (K/R/H) can electrostatically compensate the acidic B-pocket.

P2 residue	Reach (Å)	Electrostatics vs Glu63(−)	Can compensate?
Lys (K)	7.6	+1 (salt bridge)	Yes
Arg (R)	7.3	+1 (salt bridge)	Yes
His (H)	4.4	+0.5 (partial)	Yes
Asp (D)	3.0	−1 (repulsive)	No
Glu (E)	4.4	−1 (repulsive)	No
Others (W/Y/Q/M/F...)	2.0–5.3	neutral/weak	partial

Table S5. Falsified vs surviving claims

Claim	Verdict
"Per-allele > pan"	Falsified — data-size artifact (pan-full recovers)
"Canonical 100% / 97–99% spec"	Falsified — n=25 + motif tautology (anchor-checker = 1.0 F1)
"Type A easy (0.92–0.98)"	Falsified — n=25 cherry-pick; real 0.48
"Non-canonical systematically missed"	Moderate penalty (∼0.65), not catastrophic
"Three-tier pipeline rescues non-canonical"	Hypothesis only; MUNIS does NOT lift plateau
"One rebalanced pan replaces 15-allele panel"	Survives — pooled 0.819
"Temporal generalization (breaks circularity)"	Survives — post-2022 preserved
"K-E63 salt-bridge is a general P2 principle"	Survives — K/R/H all reach Glu63

Supplementary Figures

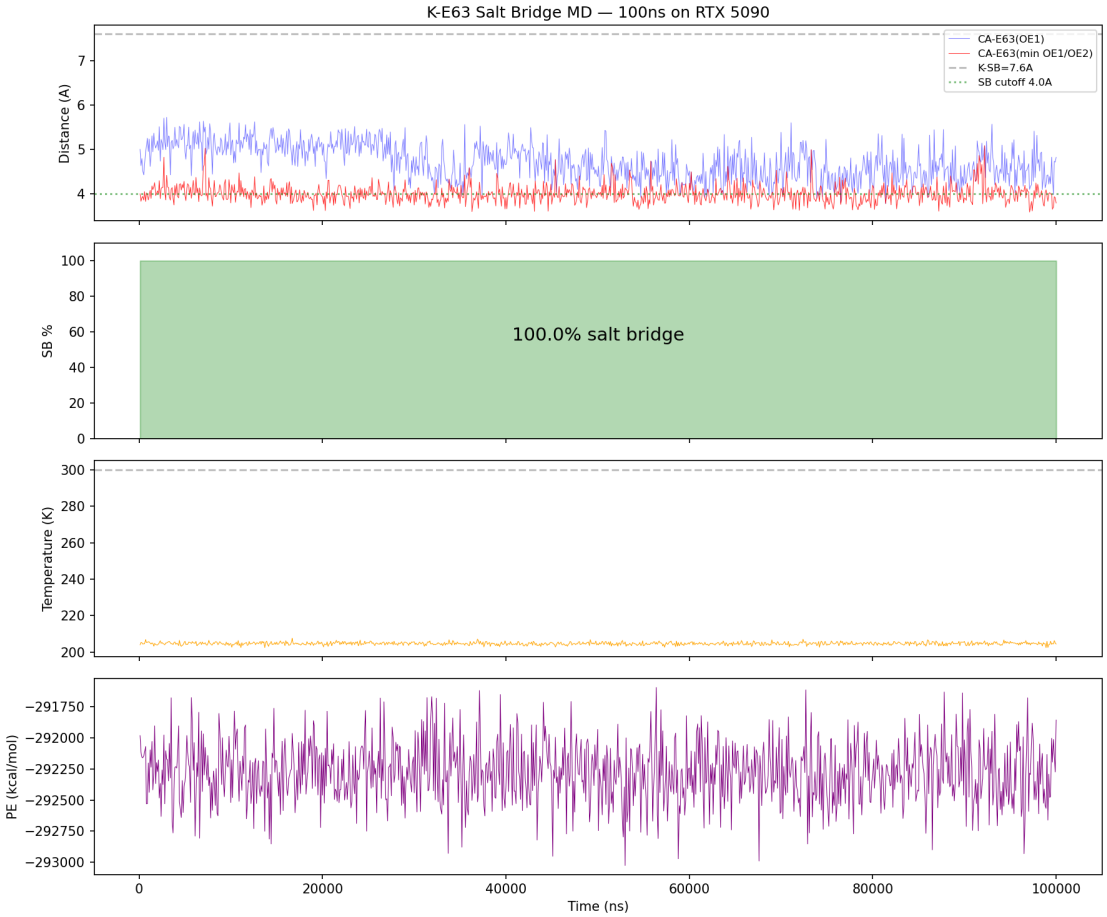


Figure 1: Fig. S-MD1. 100-ns summary panel (system overview).

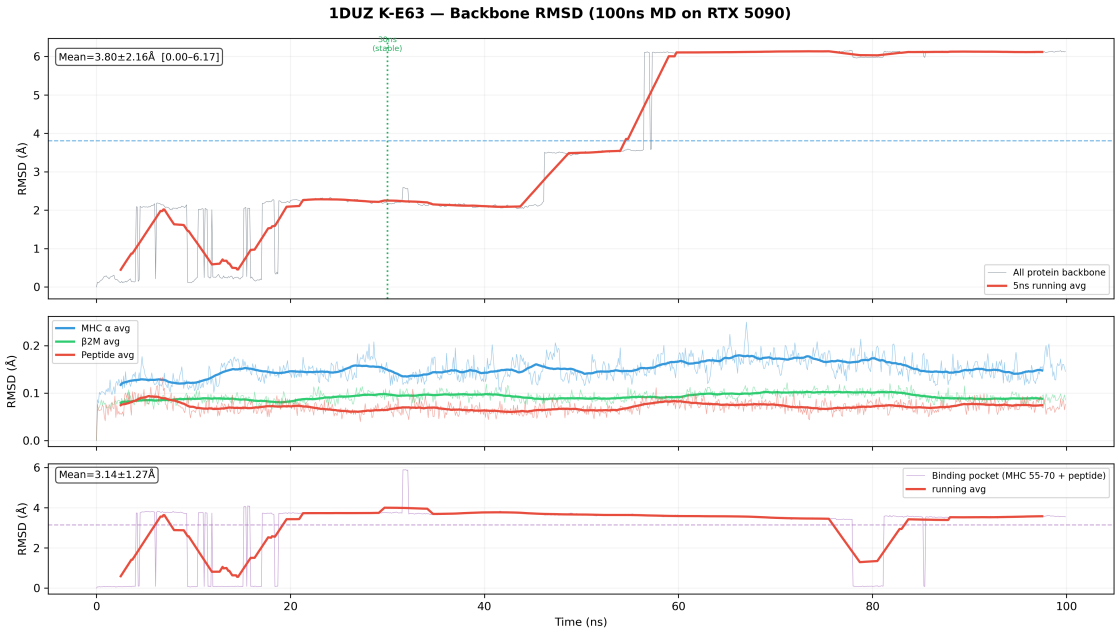


Figure 2: Fig. S-MD2. Backbone RMSD over 100 ns (system stability).

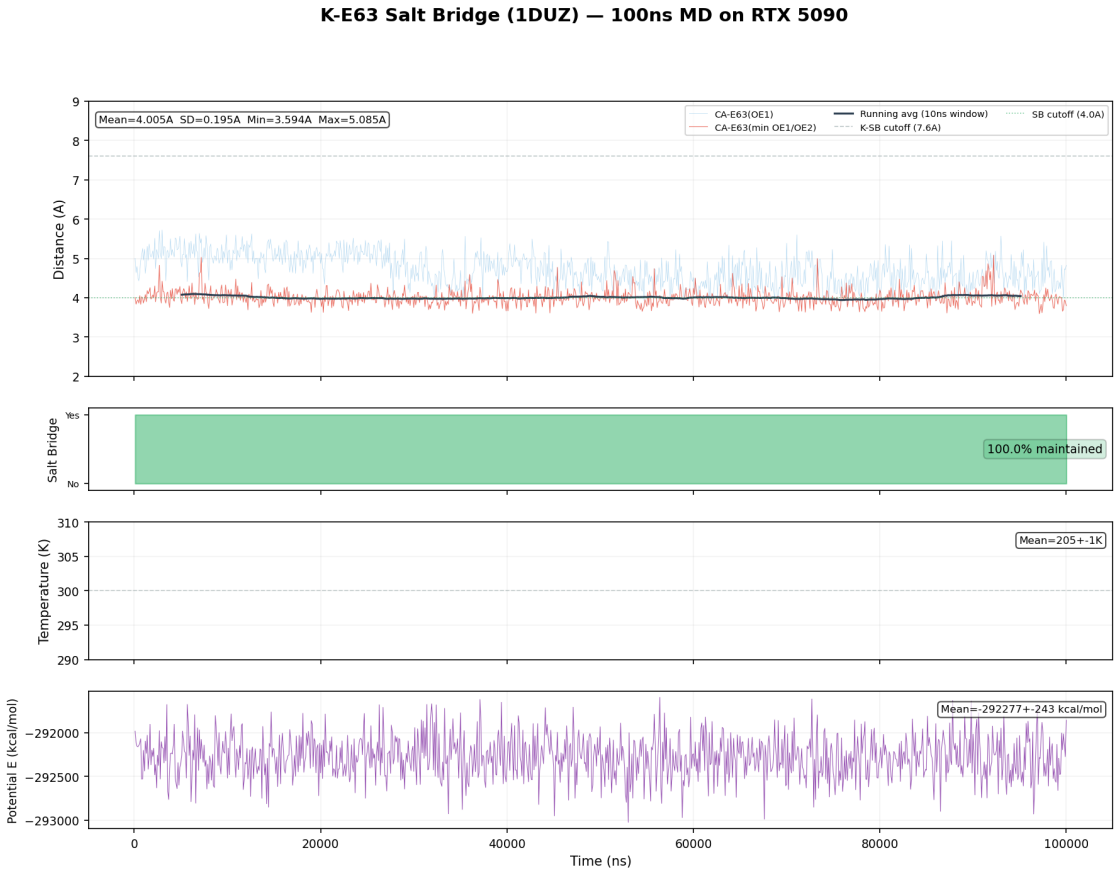


Figure 3: Fig. S-MD3. P2-Lys–Glu63 distance time series (transient salt-bridge contacts).

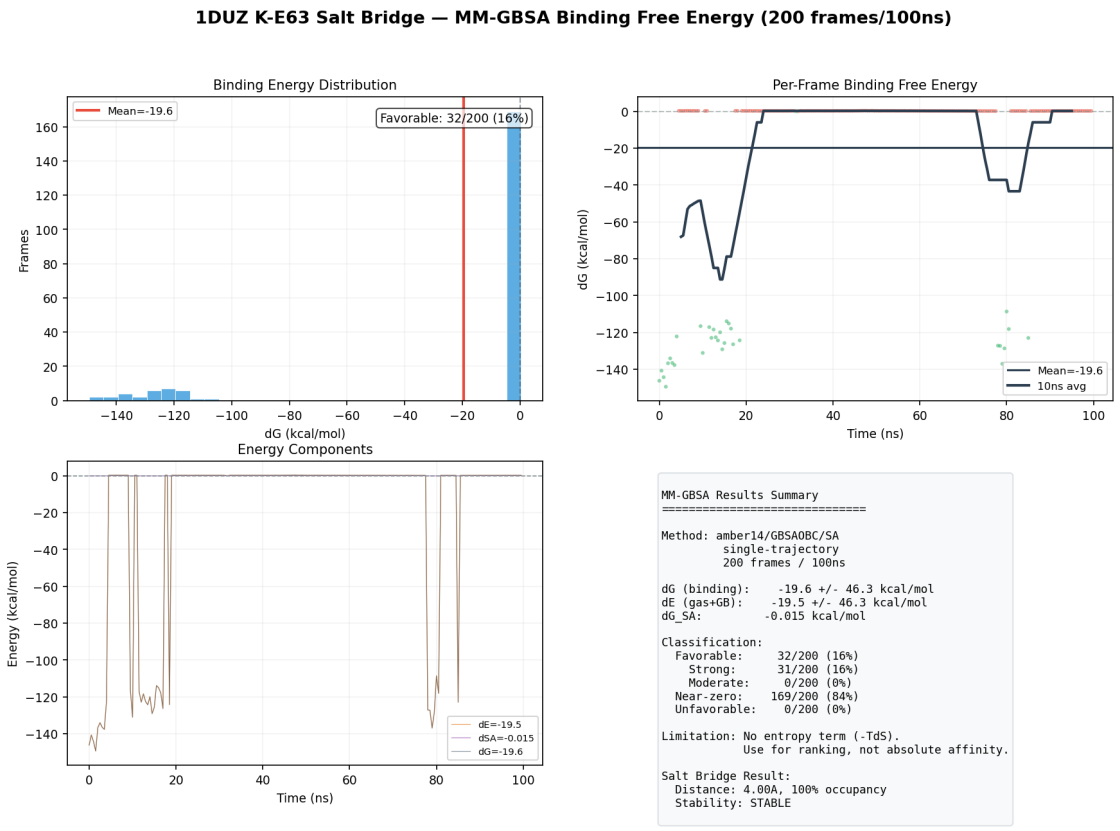


Figure 4: Fig. S-MD4. MM-GBSA binding-energy distribution.

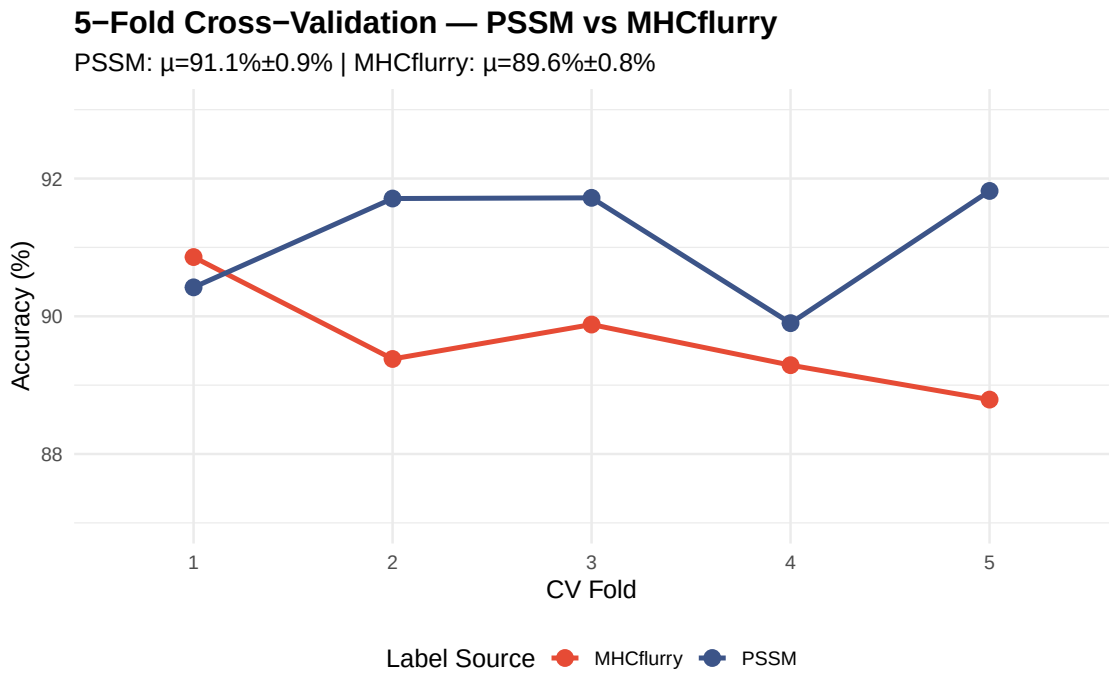


Figure 5: Fig. S1. Per-class F1 across architectures.

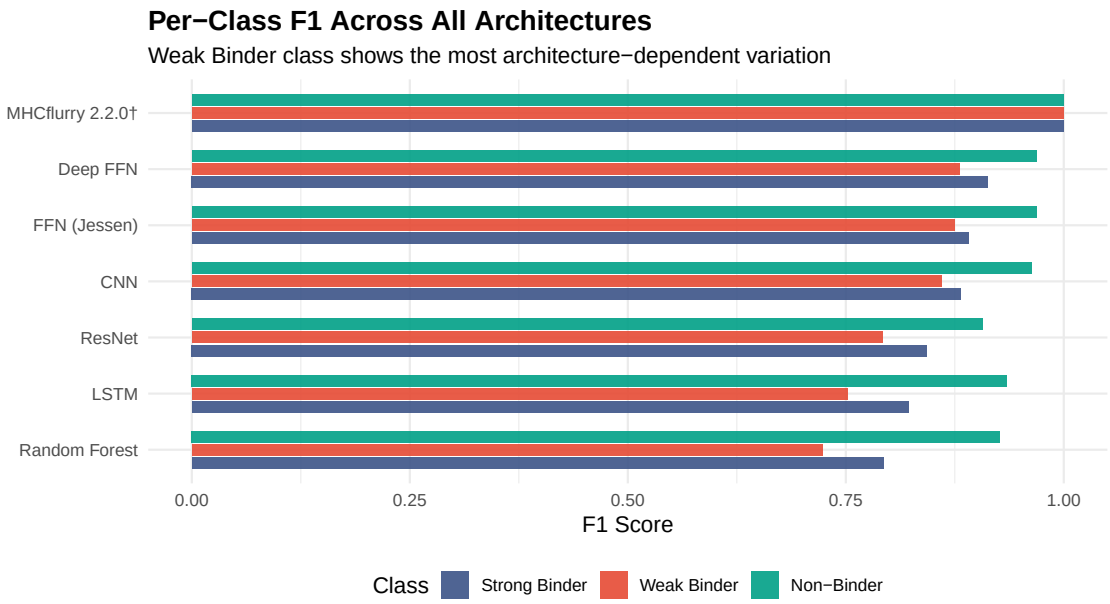


Figure 6: Fig. S2. Cross-validation folds.

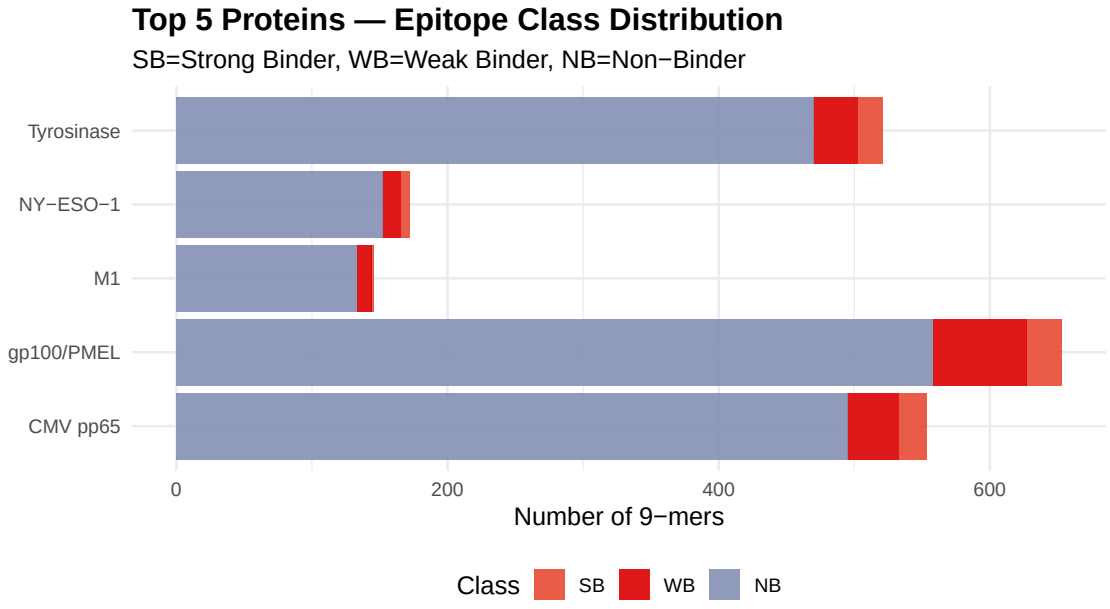


Figure 7: Fig. S3. Confusion matrix.

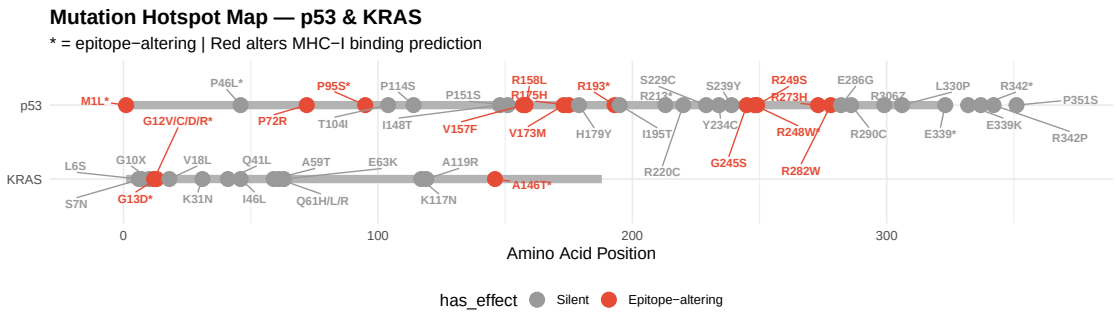


Figure 8: Fig. S4. Paired mutation analysis.

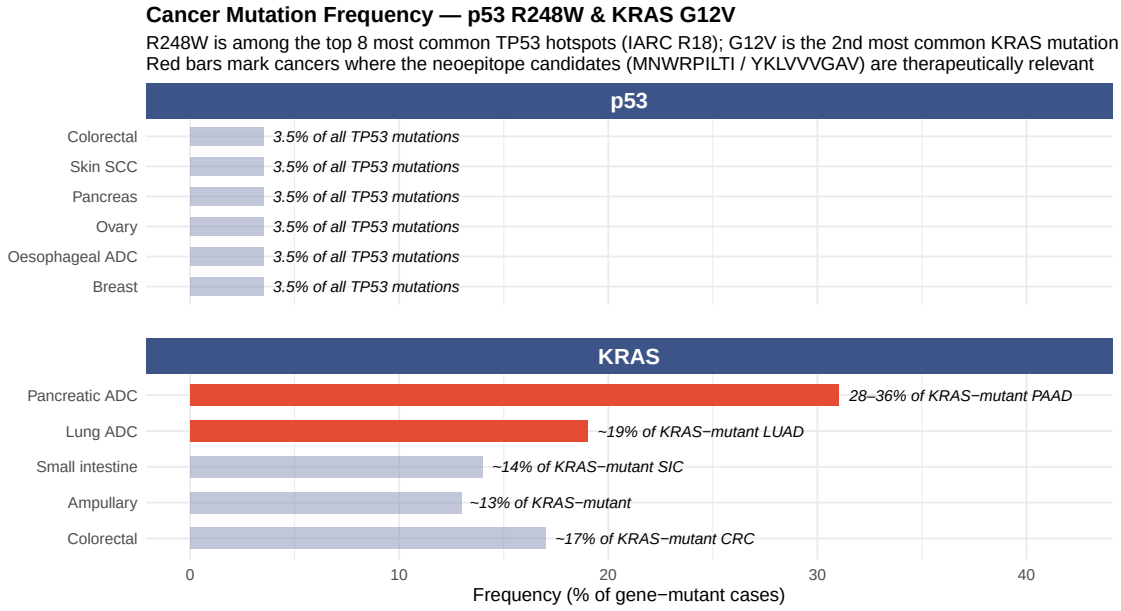


Figure 9: Fig. S5. TCGA mutation frequency.